Dart Programming Language

Operator — Operand

2 + 2

Unary

x++

--y

1) Unary Plus (+)
2) Unary Minus (-)
3) Unary Increment
   - Prefix: ++x
   - Postfix: x++
4) Decrement op
   - Prefix: --x
   - Postfix: x--
5) Logical NOT (!)
6) Bitwise NOT (~)
7) (await)

Binary

x <= y

1) Arithmetic op
   + - * / ~/ %
2) Assignment op
   = += -= *= /= ~/= %=
3) Logical op.
   && ||
4) Relational op
   ==, !=, >=, <=, >, <
5) Bitwise op
   & | ^ << >>
6) Null aware
   ?? ??= ?.
7) [ ] get, [ ]= set
8) .
9) is is! as

Ternary

cond ? ex1 : ex2;

3

Conditional op.

Special

Spread ...

...?

Cascade ..

?..

```dart
void main(){
  int x = 3;
  int y = x;
  print(y); // 3
}
```

```dart
void main(){
  int x = 4;
  int y = -x;
  print(y); // -4
}
```

increment

```dart
void main(){
  int x = 3;
  int y = 3;
  print(x++); // 3
  print(++y); // 4
}
```

Decrement

```dart
void main(){
  int x = 5;
  int y = 5;
  print(--x); //4
  print(y--); // 5
}
```

Precedence — Associativity

1) Unary Postfix — x++, y--, ( ), [ ], ., ?., ! — NONE
2) Unary Prefix — --x, ++y, await, ~x, -x — NONE
3) Multiplicative — * / ~/ % — L-R
4) Additive — + - — L-R
5) Shift — << >> — L-R
6) Bitwise AND — & — L-R
7) Bitwise XOR — ^ — L-R
8) Bitwise OR — | — L-R
9) Relational, Type test/Type cast — >= > <= < as, is, is! — NONE
10) equality — == != — NONE
11) logical AND — && — L-R
12) logical OR — || — L-R
13) if null — ?? — L-R
14) conditional — cond ? ex1 : ex2; — R-L
15) cascade — .. ?.. — L-R
16) assignment — = *= /= += -= — R-L
17) Spread — ... ...? — NONE

$$\underline{5 \times 3} - \underline{2 \div 2} \quad (L-R)$$

$$= 15 - 1$$

$$= 14$$

Print (5 >3 && 2<4)

R.OP L.OP R.OP

(true && true)

true

Print (a & b | c)

a=7 → 111

b=3 → 11

c=4 → 100

```
  111
& 011
-----
  011
| 100
-----
  111 → 7
```

Print(x<<2 & y)

x=4 → 100

y=2 → 10

```
    100
  1 0 0 0 0
& 0 0 0 1 0
-----------
 (0 0 0 0 0)
```

0